

Daily Geometry Drill #5 — Answers & Investigations

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Angle at the centre · Cyclic quadrilaterals · Tangent $\perp$ radius · Power of a point · Touching circles: join the centres.

Section A Rapid Recognition

1. In a circle with centre O , the angle at the circumference subtending arc AB is 20° . Write down the angle at the centre.

Answer: 40°

Method: The angle at the centre is twice the angle at the circumference on the same arc: $2 \times 20^\circ = 40^\circ$.

Investigate further: Where on the circle can the 20° angle be, and what does the angle become if the point moves onto the minor arc AB ?

2. AB is a diameter of a circle and P is any other point on the circle. Write down $\angle APB$.

Answer: 90°

Method: The angle at the centre is the straight angle 180° , so the angle at the circumference is half of it.

Investigate further: State and prove the converse: if $\angle APB = 90^\circ$, then P lies on the circle with diameter AB .

3. $ABCD$ is a cyclic quadrilateral with $\angle A = 70^\circ$ and $\angle B = 85^\circ$. Write down $\angle C$.

Answer: 110°

Method: Opposite angles of a cyclic quadrilateral sum to 180° , so $\angle C = 180^\circ - 70^\circ = 110^\circ$. ($\angle B$ is not needed.)

Investigate further: Find $\angle D$ as well, and check that the four angles sum to 360° .

4. The tangent at T to a circle makes an angle of 40° with the chord TA . By the alternate segment theorem, write down the angle in the alternate segment subtended by TA .

Answer: 40°

Method: The angle between a tangent and a chord equals the angle in the alternate segment.

Investigate further: Prove the alternate segment theorem from the angle in a semicircle, by taking the point in the alternate segment at the far end of the diameter through T .

5. Chords AB and CD of a circle intersect inside at P , with $PA = 2$, $PB = 6$ and $PC = 3$. Write down PD .

Answer: 4

Method: Intersecting chords: $PA \cdot PB = PC \cdot PD$, so $2 \cdot 6 = 3 \cdot PD$ and $PD = 4$.

Investigate further: Prove $PA \cdot PB = PC \cdot PD$ from a pair of similar triangles. Which angles are equal, and why?

6. A tangent from P touches the circle at T , and a secant through P cuts the circle at A then B . Given $PT = 3$ and $PA = 1$, write down PB .

Answer: 9

Method: Tangent–secant: $PT^2 = PA \cdot PB$, so $9 = 1 \cdot PB$ and $PB = 9$.

Investigate further: How long is the chord AB , and how far is its midpoint from P ?

7. Write down the length of the tangent from $(7, 0)$ to $x^2 + y^2 = 36$.

Answer: $\sqrt{13}$

Method: The radius to the point of contact is perpendicular to the tangent, so the tangent length is $\sqrt{7^2 - 6^2} = \sqrt{13}$.

Investigate further: For which points P is the tangent length to $x^2 + y^2 = 36$ equal to $\sqrt{13}$? Describe the locus.

8. Two tangents from the point P to a circle of unknown radius make an angle of 60° and $OP = 6$. Write down the length of each tangent.

Answer: $3\sqrt{3}$

Method: OP bisects the 60° angle, so each tangent makes 30° with OP and has length $6 \cos 30^\circ = 3\sqrt{3}$.

Investigate further: Find the radius of the circle too, and the area of the kite formed by P , the centre and the two points of contact.

9. The two tangents from P to the circle $x^2 + y^2 = 25$ make an angle of 120° . Write down OP .

Answer: $\frac{10\sqrt{3}}{3}$

Method: OP bisects the 120° angle, so $\sin 60^\circ = \frac{r}{OP}$ and $OP = \frac{5}{\sin 60^\circ} = \frac{10\sqrt{3}}{3}$.

Investigate further: As P moves away from the circle, the angle between the tangents shrinks. At what distance OP is it 90° ?

10. In the cyclic quadrilateral $ABCD$, $\angle BAC = 30^\circ$. Write down $\angle BDC$.

Answer: 30°

Method: $\angle BAC$ and $\angle BDC$ stand on the same arc BC , on the same side of BC , so they are equal.

Investigate further: If $\angle BAC = 30^\circ$ and $\angle CAD = 40^\circ$, find $\angle BCD$.

Section B Manipulation Drills

1. A tangent at T makes an angle of 55° with chord TA . The angle in the alternate segment subtended by TA is?

- A) 35°
- B) 55°
- C) 110°
- D) 70°

Answer: B) 55°

Method: Alternate segment theorem: the angle between the tangent and the chord equals the angle in the alternate segment. Option A is its complement; option C doubles it (that is the angle at the centre).

Investigate further: The angle at the centre on chord TA is 110° . Show this directly from the isosceles triangle OTA and the right angle between OT and the tangent.

2. $ABCD$ is cyclic with $\angle A = 2x$ and $\angle C = 3x$. Find x .

- A) 24
- B) 30
- C) 36
- D) 72

Answer: C) 36

Method: Opposite angles of a cyclic quadrilateral are supplementary: $2x + 3x = 180^\circ$, so $x = 36$.

Investigate further: Can $\angle B$ and $\angle D$ also be in the ratio 2 : 3? Find them.

3. A *tangential* (circumscribed) quadrilateral $ABCD$ has $AB = 5$, $BC = 7$ and $CD = 6$. By Pitot's theorem ($AB + CD = BC + DA$), find DA .

- A) 4
- B) 6
- C) 8
- D) 18

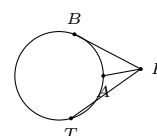
Answer: A) 4

Method: The tangent lengths from each vertex are equal, so opposite sides have equal sums: $AB + CD = BC + DA$, i.e. $5 + 6 = 7 + DA$ and $DA = 4$.

Investigate further: Prove Pitot's theorem by labelling the four pairs of equal tangent lengths.

4. From external P , tangent $PT = 12$ touches a circle and secant PAB meets it with $PA = 8$. Find the chord AB .

- A) 6
- B) 10
- C) 12
- D) 18



Answer: B) 10

Method: $PT^2 = PA \cdot PB$ gives $144 = 8 \cdot PB$, so $PB = 18$ and $AB = 18 - 8 = 10$. Option D is PB .

Investigate further: Find the distance from the centre to the chord AB if the circle has radius 13.

5. From an external point P , two secants meet the same circle at $P-A-B$ and $P-C-D$, with $PA = 4$, $PB = 18$ and $PC = 6$. Write down the length of the chord CD .

Answer: 6

Method: $PA \cdot PB = PC \cdot PD$ gives $72 = 6 \cdot PD$, so $PD = 12$ and $CD = 12 - 6 = 6$.

Investigate further: Where must a tangent from P touch, and how long is it?

6. The point P has $OP = 6$ with respect to a circle of radius 3. The angle between the two tangents from P is?

- A) 30°
- B) 45°
- C) 60°
- D) 90°

Answer: C) 60°

Method: OP bisects the angle between the tangents, and $\sin \frac{\theta}{2} = \frac{r}{OP} = \frac{1}{2}$, so $\frac{\theta}{2} = 30^\circ$ and $\theta = 60^\circ$.

Investigate further: Show that the two points of contact and P form an equilateral triangle here.

7. Two circles of radius 6 each pass through the other's centre. They meet at A and B . How long is AB ? (after *Primer 1.4 P16*)

- A) 6
- B) 12
- C) $6\sqrt{2}$
- D) $6\sqrt{3}$
- E) $3\sqrt{3}$

Answer: D) $6\sqrt{3}$

Method: The centres O_1 , O_2 and A form an equilateral triangle of side 6, and so do O_1 , O_2 and B . AB is twice the height of one of them: $2 \cdot 3\sqrt{3} = 6\sqrt{3}$. Option E is only the height.

Investigate further: Find the area of the region inside both circles. (C3 does this for radius 2.)

8. Two concentric circles have radii in the ratio 2 : 5. AC is a diameter of the larger circle, and the chord BC of the larger circle touches the smaller circle. If $AB = 8$, what is the radius of the larger circle? (after *Primer 1.4 P13*)

- A) 16
- B) 20

- C) 8
D) 12
E) 10

Answer: E) 10

Method: $\angle ABC = 90^\circ$ (angle in a semicircle). The radius OT to the point of contact is perpendicular to BC , so $OT \parallel AB$, and since O is the midpoint of AC , $OT = \frac{1}{2}AB = 4$. That is the small radius, so the large radius is $\frac{5}{2} \cdot 4 = 10$.

Investigate further: Find BC , and check that T is its midpoint.

9. The line ST is a chord of the larger of two concentric circles and touches the smaller one. If $ST = 30$, find the area of the region between the circles. (after Primer 1.4 P8)

Answer: 225π

Method: If the radii are R and r , the point of contact is the midpoint of ST , so $R^2 - r^2 = 15^2$. The area between the circles is $\pi(R^2 - r^2) = 225\pi$, whatever the radii are.

Investigate further: Why does the answer not depend on the radii? What happens as the smaller circle shrinks to a point?

10. AB is a diameter of a circle and C is a point on the circle with $\angle ABC = 55^\circ$. Write down $\angle BAC$.

Answer: 35°

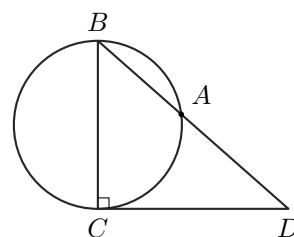
Method: $\angle ACB = 90^\circ$ (angle in a semicircle), so $\angle BAC = 180^\circ - 90^\circ - 55^\circ = 35^\circ$.

Investigate further: Find the angle between the tangent at A and the chord AC .

Section C Substitution & Structure

1. BC is a diameter of a circle and CD is the tangent at C . The line BD meets the circle again at A . If $AB = 7$ and $AD = 9$, what is the radius of the circle? (after Primer 1.4 P40)

- A) 12
B) $4\sqrt{7}$
C) $2\sqrt{7}$
D) 8
E) 6



Answer: C) $2\sqrt{7}$

Method: Tangent-secant from D : $DC^2 = DA \cdot DB = 9 \cdot 16 = 144$, so $DC = 12$. The tangent is perpendicular to the diameter, so $BC^2 = BD^2 - DC^2 = 256 - 144 = 112$, $BC = 4\sqrt{7}$ and the radius is $2\sqrt{7}$. Option A is DC ; option B is the diameter.

Investigate further: Find AC in two ways: from triangle ABC (right-angled at A) and from triangle ACD .

2. A regular octagon is inscribed in a circle of radius 2, and X is any point on the circle. What is

the sum of the squares of the distances from X to the eight vertices? (after SMC 2025 Q21)

- A) 64
- B) 32
- C) 128
- D) 48
- E) 16

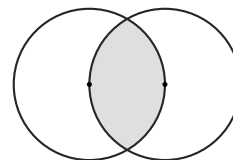
Answer: A) 64

Method: Put the centre at the origin. For a vertex V , $XV^2 = |X|^2 + |V|^2 - 2X \cdot V = 8 - 2X \cdot V$. The eight vertex vectors sum to zero, so the dot products cancel and the total is $8 \cdot 8 = 64$, for every X on the circle. Option B forgets the $|X|^2$ terms.

Investigate further: Show that the answer $2nR^2$ holds for any regular n -gon and any X on the circle. What is the sum if X is the centre?

3. Two circles of radius 2 each pass through the other's centre. What is the area of the shaded region common to both circles? (after Primer 1.4 P24)

- A) $\frac{4\pi}{3} - \sqrt{3}$
- B) $\frac{8\pi}{3} - \sqrt{3}$
- C) $\frac{4\pi}{3} + 2\sqrt{3}$
- D) $\frac{8\pi}{3} - 2\sqrt{3}$
- E) $2\pi - 2\sqrt{3}$



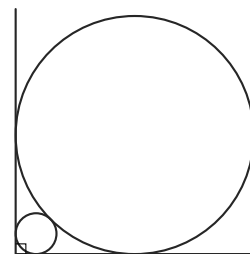
Answer: D) $\frac{8\pi}{3} - 2\sqrt{3}$

Method: The common chord and the two centres form two equilateral triangles, so each centre sees the chord at 120° . The region is two segments, each a 120° sector minus a triangle: $\frac{1}{3}\pi \cdot 4 - \frac{1}{2} \cdot 4 \sin 120^\circ = \frac{4\pi}{3} - \sqrt{3}$. Two of them: $\frac{8\pi}{3} - 2\sqrt{3}$. Option A is one segment.

Investigate further: What fraction of one circle is the common region? Show that it is the same for every radius.

4. A circle of radius 1 touches both arms of a right angle. A larger circle touches both arms and the first circle. What is the radius of the larger circle? (after Primer 1.4 P35)

- A) $1 + \sqrt{2}$
- B) $2 + \sqrt{2}$
- C) $2\sqrt{2}$
- D) 3
- E) $3 + 2\sqrt{2}$



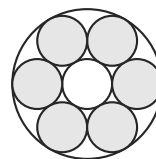
Answer: E) $3 + 2\sqrt{2}$

Method: Both centres lie on the bisector of the angle, at distances $\sqrt{2}$ and $R\sqrt{2}$ from the vertex. The circles touch, so $R\sqrt{2} - \sqrt{2} = R + 1$, giving $R = \frac{\sqrt{2}+1}{\sqrt{2}-1} = (\sqrt{2} + 1)^2 = 3 + 2\sqrt{2}$.

Investigate further: For an angle 2α show the ratio of consecutive radii is $\frac{1+\sin\alpha}{1-\sin\alpha}$. For which angle is it exactly 3?

5. Six equal discs lie between two concentric circles. Each disc touches its two neighbours and both circles. The inner circle has radius 1. What is the radius of the outer circle? (after SMC 2016 Q21)

- A) 3
 B) $2 + \sqrt{3}$
 C) $1 + \sqrt{3}$
 D) $2\sqrt{3}$
 E) 4



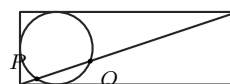
Answer: A) 3

Method: Let the discs have radius ρ . Their centres lie on a circle of radius $1 + \rho$, and neighbouring centres are 2ρ apart, which is the side of a regular hexagon inscribed in that circle: $2\rho = 1 + \rho$. So $\rho = 1$ and the outer radius is $1 + 2\rho = 3$.

Investigate further: Repeat with n discs: show $\rho = \frac{\sin(180^\circ/n)}{1 - \sin(180^\circ/n)}$. For which n are the discs larger than the inner circle?

6. A circle of radius 1 touches three sides of a 6×2 rectangle. A diagonal of the rectangle cuts the circle at P and Q . What is the length PQ ? (after SMC 2017 Q19)

- A) $\frac{4\sqrt{5}}{5}$
 B) $\frac{2\sqrt{15}}{5}$
 C) $\sqrt{3}$
 D) $\frac{6}{5}$
 E) $\frac{2\sqrt{10}}{5}$



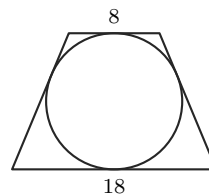
Answer: B) $\frac{2\sqrt{15}}{5}$

Method: Put the rectangle at $0 \leq x \leq 6$, $0 \leq y \leq 2$; the circle has centre $(1, 1)$. The diagonal is $x - 3y = 0$, at distance $\frac{|1-3|}{\sqrt{10}} = \frac{2}{\sqrt{10}}$ from the centre. The half-chord is $\sqrt{1 - \frac{4}{10}} = \sqrt{\frac{3}{5}}$, so $PQ = 2\sqrt{\frac{3}{5}} = \frac{2\sqrt{15}}{5}$. Option A is the answer for a 4×2 rectangle.

Investigate further: How long must the rectangle be for the diagonal to miss the circle altogether? For it to just touch?

7. An isosceles trapezium has parallel sides of lengths 8 and 18, and all four of its sides touch a circle. What is the radius of the circle? (after SMC 2012 Q20)

- A) $\frac{13}{2}$
 B) 5
 C) 12
 D) 6
 E) $4\sqrt{3}$



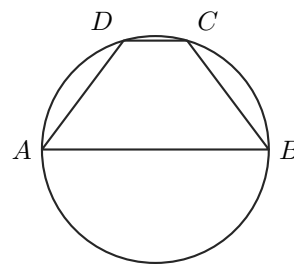
Answer: D) 6

Method: A quadrilateral with an incircle has equal sums of opposite sides (Pitot): $8 + 18 = 2\ell$, so each slanted side is $\ell = 13$. Each overhangs by $\frac{18-8}{2} = 5$, so the height is $\sqrt{13^2 - 5^2} = 12$. The circle touches both parallel sides, so its diameter is the height and the radius is 6. Option C is the height.

Investigate further: Show that for any isosceles trapezium with an incircle, the radius is $\frac{1}{2}\sqrt{ab}$ where a and b are the parallel sides.

8. $ABCD$ is a cyclic quadrilateral in which AB is a diameter of the circle, $AB = 25$ and $AD = BC = 15$. What is the length CD ? (after Primer 1.4 P33)

- A) 10
 B) 15
 C) 13
 D) 9
 E) 7



Answer: E) 7

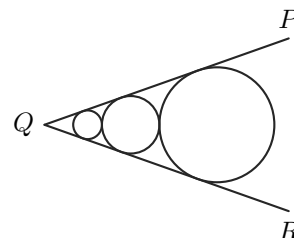
Method: $\angle ADB = \angle ACB = 90^\circ$, so $BD = AC = \sqrt{25^2 - 15^2} = 20$. Ptolemy's theorem for $ABCD$: $AC \cdot BD = AB \cdot CD + AD \cdot BC$, i.e. $400 = 25CD + 225$, so $CD = 7$. Without Ptolemy: D is $\frac{15^2}{25} = 9$ along AB from A , and C is 9 from B , so $CD = 25 - 18 = 7$.

Investigate further: Keep $AB = 25$ and let $AD = BC = x$. For which x does the trapezium become a triangle ($C = D$)?

Section D Challenge

1. Three circles each touch both arms QP and QR of an angle, and each touches the next. The largest radius is 4 times the smallest. What is $\cos \angle PQR$? (after SMC 2011 Q24)

- A) $\frac{7}{9}$
 B) $\frac{1}{3}$
 C) $\frac{2}{3}$
 D) $\frac{8}{9}$
 E) $\frac{4\sqrt{2}}{9}$



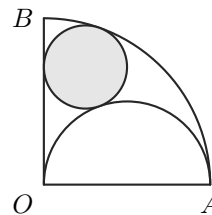
Answer: A) $\frac{7}{9}$

Method: The centres lie on the bisector; a circle whose centre is d from Q has radius $d \sin \alpha$, where $2\alpha = \angle PQR$. Consecutive circles touch, so their radii are in a constant ratio k with $\frac{k-1}{k+1} = \sin \alpha$. Three circles give ratio $k^2 = 4$, so $k = 2$ and $\sin \alpha = \frac{1}{3}$. Then $\cos 2\alpha = 1 - 2\sin^2 \alpha = \frac{7}{9}$. Option B is $\sin \alpha$; option E is $\sin 2\alpha$.

Investigate further: Continue the chain towards Q for ever. Show that the radii form a geometric sequence, and find the total area of all the new circles in terms of the smallest original radius r .

2. OAB is a quarter circle with radius 4. A semicircle is drawn on OA as diameter, inside the quarter circle. A circle touches OB , the semicircle and the arc AB . What is its radius? (after SMC 2014 Q19)

- A) $\frac{4}{3}$
 B) 1
 C) $\frac{2}{3}$
 D) $\sqrt{2}$
 E) $\frac{3}{4}$



Answer: B) 1

Method: Put O at the origin with OA along the x -axis. The circle touches OB (the y -axis), so its centre is (r, y) . It touches the semicircle (centre $(2, 0)$, radius 2) externally: $(r - 2)^2 + y^2 = (r + 2)^2$, so $y^2 = 8r$. It touches the arc internally: $r^2 + y^2 = (4 - r)^2$, so $y^2 = 16 - 8r$. Hence $8r = 16 - 8r$ and $r = 1$.

Investigate further: Replace the quarter circle's radius 4 by $2a$ and the semicircle by one of radius a . Show that the answer is always $\frac{a}{2}$.

3. A triangle with angles 30° , 45° and 105° is inscribed in a circle of radius 2. What is its area? (after SMC 2021 Q22)

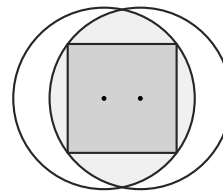
- A) $2 + \sqrt{3}$
 B) $2\sqrt{3}$
 C) $1 + \sqrt{3}$
 D) $3 + \sqrt{3}$
 E) 2

Answer: C) $1 + \sqrt{3}$

Method: By the extended sine rule each side is $2R \sin(\text{opposite angle})$: $a = 4 \sin 30^\circ = 2$, $b = 4 \sin 45^\circ = 2\sqrt{2}$. The area is $\frac{1}{2}ab \sin C = \frac{1}{2} \cdot 2 \cdot 2\sqrt{2} \cdot \sin 105^\circ$, and $\sin 105^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$, giving $\frac{\sqrt{2}(\sqrt{6} + \sqrt{2})}{2} = \sqrt{3} + 1$.

Investigate further: Prove that $[ABC] = 2R^2 \sin A \sin B \sin C$ for every triangle inscribed in a circle of radius R . Which inscribed triangle has the largest area?

4. Two circles of radius 5 have centres 2 apart. A square is inscribed in the region common to both circles, with its sides parallel and perpendicular to the line of centres. What is the area of the square? (after SMC 2019 Q25)



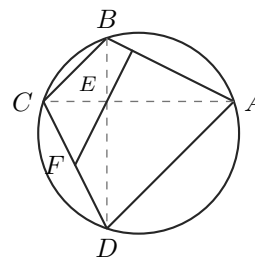
- A) 64
B) 32
C) 25
D) 36
E) 48

Answer: D) 36

Method: Put the centres at $(\pm 1, 0)$. By symmetry the square is centred at the origin with corners $(\pm \frac{s}{2}, \pm \frac{s}{2})$, and the corner $(\frac{s}{2}, \frac{s}{2})$ lies on the circle centred at $(-1, 0)$: $(\frac{s}{2} + 1)^2 + \frac{s^2}{4} = 25$, i.e. $s^2 + 2s - 48 = 0$, so $(s + 8)(s - 6) = 0$, $s = 6$ and the area is 36. Option A takes the square across the full width of the region ($s = 8$), which leaves its corners outside.

Investigate further: The region is 8 wide and $4\sqrt{6}$ tall. Find the largest rectangle with sides parallel to the axes that fits inside it.

5. $ABCD$ is a cyclic quadrilateral whose diagonals are perpendicular and meet at E . The line through E perpendicular to AB meets CD at F . Prove that F is the midpoint of CD .



Answer: Proof: see method

Method: Let the line meet AB at H , so $EH \perp AB$. In triangle AEB , right-angled at E , the altitude EH gives $\angle HEA = 90^\circ - \angle EAB = \angle ABE$. Now $\angle ABE = \angle ABD = \angle ACD$ (angles on the arc AD), and $\angle HEA = \angle FEC$ (vertically opposite). So $\angle FEC = \angle FCE$ and $FE = FC$. Triangle CED is right-angled at E , so $\angle FED = 90^\circ - \angle FEC = 90^\circ - \angle FCE = \angle FDE$, and $FE = FD$. Hence $FC = FE = FD$: F is the midpoint of CD . ■

Investigate further: This is Brahmagupta's theorem. Use it to show that the distance from the circle's centre to AB is half of CD .

Top 5 Patterns — Worksheet #5

- Angles on an arc.** The angle at the centre is twice the angle at the circumference; angles on the same arc are equal; opposite angles of a cyclic quadrilateral sum to 180° . (see A1, A2, A3, A10, B2, B10)
- Tangent $\perp$ radius.** It gives the right angle behind tangent lengths, the angle between tangents and the alternate segment theorem. (see A4, A7, A8, A9, B1, B6)
- Power of a point.** $PA \cdot PB$ is the same for every line through P that meets the circle, and equals PT^2 for a tangent. (see A5, A6, B4, B5, B9, C1)
- Touching circles: join the centres.** The distance is $r_1 + r_2$ (or $|r_1 - r_2|$), and circles in an angle have centres on its bisector with radius proportional to distance. (see C4, C5, D1, D2)

- 5. Chords, segments and lenses.** Use the perpendicular from the centre to a chord, sector minus triangle for a segment, and Ptolemy for cyclic quadrilaterals. (*see B7, B8, C3, C6, C8, D4*)

Common Traps — Worksheet #5

- 1. Quoting the angle at the centre for the angle at the circumference.** Halve it (or double it) once, in the right direction. (*see A1, B1*)
- 2. Stopping at the tangent length.** In C1 the tangent is only the first step; the question asks for the radius. (*see B4, C1*)
- 3. Using the height or diameter where the radius is asked.** 12 in C7 and $4\sqrt{7}$ in C1 are one step short. (*see C1, C7, B8*)
- 4. Mixing up external and internal tangency.** Circles touching externally have centres $r_1 + r_2$ apart; a circle touching a larger one from inside has centres $R - r$ apart, as with the arc in D2 and the outer circle in C5. (*see C4, C5, D2*)
- 5. Counting one segment for a lens, or the whole width for an inscribed square.** Check that every corner of your shape is actually on the boundary. (*see C3, D4, D3*)

“With circles, draw what you cannot see: the centres, the radii to the points of contact, and the line joining two centres.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths